Started on	2024-03-14 06:54
State	Finished
Completed on	2024-03-14 08:26
Time taken	1 hour 32 mins
Grade	**18.33** out of 20.00 (**92**%)

Question **1**
Correct
Mark 2.00 out of 2.00

Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$.

Consider a function $f : A \to B$ defined by $f(a) = 2$, $f(b) = 1$, $f(c) = 4$, $f(d) = 1$ and $f(e) = 1$.

Then

a. image of "a" is 2 ✓

b. preimage of 1 is $\{$ b,d,e ✓ $\}$

c. domain is $\{$ a,b,c,d,e ✓ $\}$

d. codomain is $\{$ 1,2,3,4 ✓ $\}$

e. range is $\{$ 1,2,4 ✓ $\}$

Question **2**
Partially correct
Mark 1.33 out of 2.00

Determine whether f is a function from $\mathbb{Z}$ to $\mathbb{R}$.

a. If $f(n) = \pm n$, then f is not a function ✓

b. If $f(n) = \sqrt{n^2+1}$, then f is a function ✓

c. If $f(n) = \frac{1}{n+1}$, then f is a function ✗

Question **3**
Correct
Mark 2.00 out of 2.00

Consider the functions from the set of students in this class. Is the function one-to-one if it assigns to a student his or her

a. mobile phone number one-to-one ✓

b. student ID one-to-one ✓

c. final grade in DM not one-to-one ✓

d. home town not one-to-one ✓

Question 4
Partially correct
Mark 1.50 out of 2.00

For each function on $\mathbb{R}$ below decide if it is onto or not.

a. $|x|$ onto ✗

b. x onto ✓

c. x^2 not onto ✓

d. x^3 onto ✓

Question 5
Correct
Mark 1.00 out of 1.00

We know that to convert Celsius to Fahrenheit we use formula

$F = 1.8 \times C + 32$

Choose a formula for backward conversion Fahrenheit to Celsius.

Select one:

- $C = 1.8F + 32$
- $C = \frac{1}{1.8}F - 32$
- $C = \frac{F-32}{1.8}$ ✓
- $C = 1.8(F - 32)$

Your answer is correct.

The correct answer is: $C = \frac{F-32}{1.8}$

Question 6
Complete
Mark 2.00 out of 2.00

Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 1$ are functions from $\mathbb{R}$ to $\mathbb{R}$.

Write your solution, take a picture and attach it below.

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Comment:

Question 7
Correct
Mark 1.00 out of 1.00

Match the corresponding values:

$\lfloor 1.1 \rfloor$	1 ✓
$\lfloor -0.1 \rfloor$	-1 ✓
$\lceil 1.1 \rceil$	2 ✓
$\lceil -0.1 \rceil$	0 ✓

Your answer is correct.

The correct answer is: $\lfloor 1.1 \rfloor$ → 1, $\lfloor -0.1 \rfloor$ → -1, $\lceil 1.1 \rceil$ → 2, $\lceil -0.1 \rceil$ → 0

Question 8
Complete
Mark 1.00 out of 1.00

Construct a bijection between the sets $\{0, 1, 2, 3, 4\}$ and $\{2, 4, 6, 8, 10\}$.

Write your answer, take a picture and attach it below.

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Comment:

Question 9
Complete
Mark 2.00 out of 2.00

i. Construct a one-to-one function from the set $B = \{red, green, blue, yellow\}$ to the set A of letters in English alphabet.

ii. Is your conclusion that $|A| \leq |B|$ or $|B| \leq |A|$?

Write your answers, take a picture and attach it below.

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Comment:

Question 10
Complete
Mark 2.50 out of 3.00

Show that the set of all multiples of 10 is countable.

Write your proof, take a picture and attach it below.

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Comment:

Question 11

Correct

Mark 2.00 out of 2.00

Determine whether each of these sets is finite, countably infinite or uncountable.

a. the even negative integers [countably infinite] ✓

b. the integers with absolute value less than 1000 [finite] ✓

c. the real numbers between 2 and 3 [uncountable] ✓